Arnav A. Dhariya

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Education

University of California, Irvine Expected June 2027
B.S. in Computer Science, Specialization in Intelligent Systems *GPA:* ***3.92***

- **Relevant Coursework:** Machine Learning, Artificial Intelligence, Statistics for ML, Linear Algebra, Data Structures & Algorithms, Databases, Usable Security and Privacy (Graduate), Operating Systems

Technical Skills

Languages & Libraries: Python (pandas, NumPy, scikit-learn, PyTorch), SQL, C++, Git
LLMs & Inference: vLLM, Qwen, KV-cache optimization, Claude, Codex
Profiling & Infrastructure: Modal, NVIDIA Nsight Systems, PyTorch Profiler
Data & Optimization: PostgreSQL, MySQL, FAISS, Gurobi, differential privacy

Experience

Full Stack Data Lab, Carnegie Mellon University Jul 2026 – Present
Undergraduate Researcher (Advisor: ***Prof. Shreya Shankar****)*

- Co-designed and implemented Quail, an execution engine for AI-native SQL operators on vLLM that combines prefix/KV-cache reuse with recurrent-state checkpointing for hybrid models, achieving **3–4x** lower latency than default vLLM at identical accuracy across Qwen3, Qwen3.5, Qwen3.8, and LFM models
- Profiled a custom vLLM benchmark with NVIDIA Nsight Systems and PyTorch Profiler on Modal, tracing kernel-level and memory behavior to find that semantic operators saturate the KV cache and trigger frequent evictions, and used this diagnosis to drive Quail's design
- Authored cost-model analyses for AI filters (blog) establishing an ideal-cost baseline that no system, including Quail, has yet reached; designing a follow-on optimization to close the gap

Information Systems Group, University of California, Irvine Sep 2025 – Present
Undergraduate Researcher (Mentor: ***Dr. Vishal Chakraborty****; Advisor:* ***Prof. Sharad Mehrotra****)*

- Designed and implemented differential-privacy-based deletion mechanisms (Python, MySQL) that make deletions inference-aware: removing a record also removes correlated data from which it could be re-inferred, which standard `DELETE` does not do
- Cut collateral deletions (extra tuples removed beyond the target record) by **60–70%** for a **20-point** rise in re-inference probability, using the Exponential mechanism with a greedy Gumbel-max sampling trick; extends Pre-Insertion Post-Erasure Equivalence
- Co-authored a SIGMOD Workshop SeQureDB 2026 paper and a **SIGMOD 2027** submission; presented at UCI UROP Symposium 2026 and NorCal DB Day 2026. Currently extending the framework to **physical-layer** and **pipeline** deletions

NetSAIL, University of California, Irvine Sep 2025 – Present
Undergraduate Researcher (Mentors: ***Kapil Agrawal, Prof. Sangeetha Abdu Jyothi****)*

- Built a discrete-event simulation framework modeling large-scale vector-index sharding (FAISS IVF, BigANN, BatANN) to study load imbalance, failure propagation, and graceful degradation
- Identified problems in sharding as used in industry-standard systems and developed a heuristic to address them

Wayne Hayes Lab, University of California, Irvine Sep 2024 – Sep 2025
Undergraduate Researcher

- Optimized haplotype assembly (C++ modifications to the SAHap simulated-annealing algorithm for the Weighted Minimum Error Correction problem), cutting runtime by **20%** on datasets with **500+** SNPs
- Refined simulated annealing and genome-read prioritization to improve haplotype reconstruction accuracy across multiple SNP datasets

Publications

- V. Chakraborty, Y. Kaminsky, **A. A. Dhariya**, S. Mehrotra, F. Naumann, S. Pandey. *Inference-Aware & Privacy-Preserving Deletion in Databases.* SIGMOD Workshop on Data Management, Privacy, and Security. ACM, 2026. arXiv:2604.00326
- H. Freedman, M. Foletta, G. Fernandez, V. Sokolova, **A. A. Dhariya**, A. van der Hoek, B. Tomlinson. *A Systematic Literature Review of Climate Econometrics Research.* SSRN, April 2025. ssrn.com/abstract=6178246

Awards

- **Chancellor's Award for Excellence in Undergraduate Research**, 2025 (**top 1%** of UC undergraduate researchers)
- **Undergraduate Research Symposium Finalist** (2025, 2026)
- **Summer Undergraduate Research Program (SURP) Fellowship** (**$1,500**)
- **Dean's Honor List** (**6 quarters**)
- Selected for the **ICS Honors Program** (**selection-based**); Honors Thesis on Deletion